Visualization Tool for Electric Vehicle Charge and Range Analysis

Project Documentation

Team Members:

Siddhant Dwivedi

Vivek kumar

Vivek kumar kasana

Sumit Rajput

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1. Introduction

This project focuses on analyzing key factors related to **electric vehicle (EV) performance**, particularly **battery charge capacity and driving range**, using data visualization techniques. The goal of the project is to explore relationships between various EV specifications such as battery capacity, charging time, efficiency, and driving range.

The analysis was performed using **Tableau dashboards**, which allow interactive exploration of EV data. The dashboards are integrated into a **Flask-based web application**, providing users with an accessible and user-friendly interface to visualize and understand EV performance metrics.

This project aims to help users, researchers, and policymakers better understand the **current EV landscape and technological trends**.

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2. Problem Statement

Electric vehicles are becoming increasingly important in the transition toward **sustainable transportation**. However, many users still face challenges in understanding how factors like **battery capacity, charging infrastructure, and efficiency affect driving range**.

Consumers often struggle to compare different EV models and evaluate their performance due to scattered or complex data sources.

This project addresses the problem by analyzing EV datasets and presenting the information through **interactive visual dashboards**, enabling users to easily explore the relationships between **battery capacity, charging time, efficiency, and vehicle range**.

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3. Data Collection

The dataset used in this project contains multiple attributes related to **electric vehicle specifications and performance metrics**.

These include:

- Vehicle model name
- Battery capacity (kWh)
- Driving range (km)
- Charging time
- Efficiency (km/kWh)
- Vehicle price and category
- Charging infrastructure availability

The data was collected from publicly available EV datasets and online sources related to electric vehicle specifications and charging infrastructure.

After collection, the dataset was organized into a structured format suitable for analysis and visualization.

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4. Data Cleaning and Preparation

Before performing visualization, the dataset was carefully examined for **missing values, inconsistencies, and formatting issues**.

The following preprocessing steps were performed:

- Removing duplicate entries
- Handling missing values
- Standardizing measurement units (e.g., kWh, km, hours)
- Organizing categorical variables such as vehicle type and manufacturer
- Removing irrelevant attributes

These steps ensured that the dataset was **clean, reliable, and suitable for generating meaningful visualizations** in Tableau.

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5. Data Visualization using Tableau

Multiple dashboards were created using **Tableau** to analyze and visualize relationships between EV performance factors.

The dashboards include visualizations such as:

- **Battery Capacity vs Driving Range Analysis**

- **Charging Time Comparison between EV Models**
- **Efficiency Analysis (km per kWh)**
- **EV Model Performance Comparison**
- **Charging Infrastructure Distribution**

These visualizations allow users to interact with the data, apply filters, and explore patterns that influence EV performance and adoption.

The dashboards are embedded into a **Flask web application**, enabling users to access the visualizations through a simple and responsive web interface.

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6. Conclusion

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** demonstrates how data visualization techniques can transform complex EV datasets into clear and actionable insights.

By combining **Tableau dashboards with a Flask-based web application**, the project provides an intuitive platform for analyzing EV performance metrics.

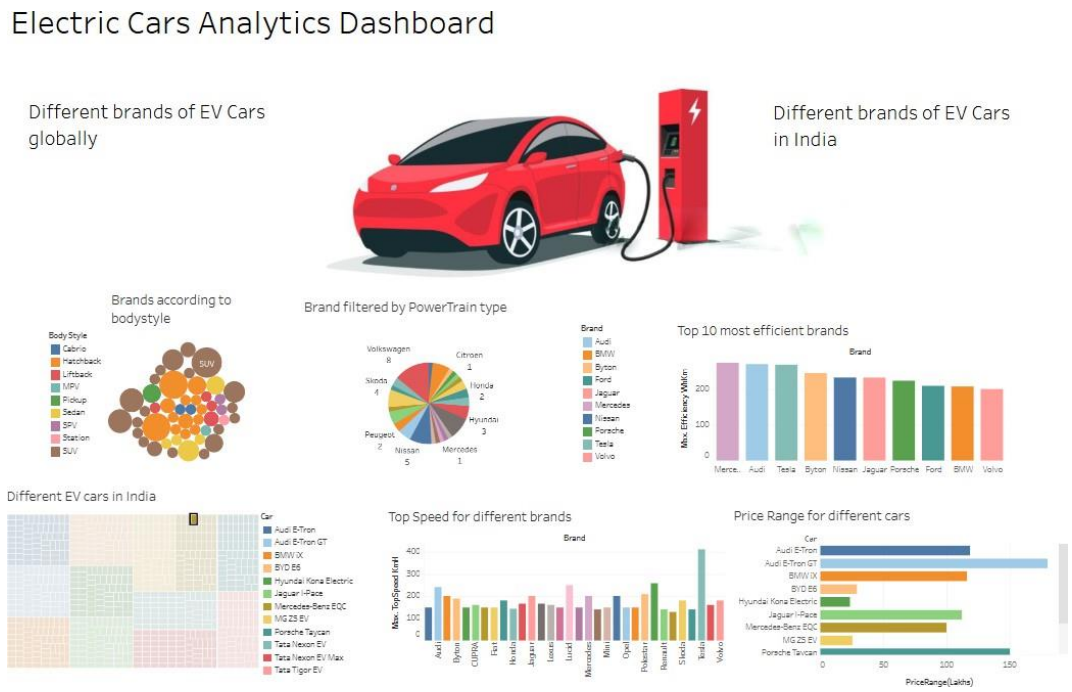
The project highlights important insights related to **battery performance, charging infrastructure, and vehicle efficiency**, which can help users make informed decisions and contribute to a better understanding of the evolving electric vehicle ecosystem.

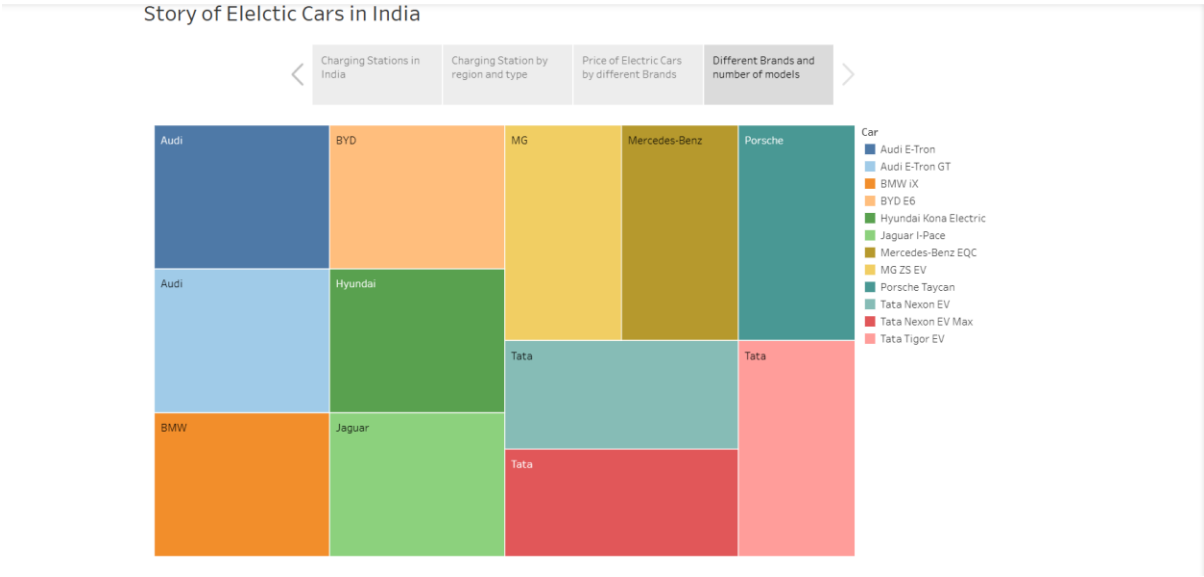
Tableau Dashboard

Dashboard – Electric Cars Analytics Dashboard

This dashboard provides a comprehensive visualization of electric vehicle data, including **different EV brands globally and in India, vehicle body styles, brand distribution by powertrain type, and the top 10 most efficient EV brands**. It also presents insights on **EV models available in India, top speed comparisons between different brands, and price ranges of various electric vehicles**.

The dashboard helps users easily analyze and compare **electric vehicle performance, efficiency, and market availability**, enabling better understanding of the EV ecosystem.





Dashboard:

Link:https://public.tableau.com/views/Book2_17729068481180/Dashboard1?:language=en-

US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Key Insights

• Battery capacity has a strong impact on the **driving range of electric vehicles**. Vehicles with higher battery capacity generally provide longer travel distances.

• Different EV brands show **variations in efficiency**, indicating differences in battery technology and vehicle design.

• The analysis of EV models available in India highlights the **growing presence of electric vehicles in the Indian market**.

• Efficiency comparison among brands shows that some manufacturers focus more on **energy-efficient vehicles with better km/kWh performance**.

• The comparison of top speeds across brands shows that electric vehicles are now **competitive with traditional internal combustion engine vehicles**.

• Price range analysis indicates that EV models are available across **different market segments, from mid-range to premium vehicles**.

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7. Website Development

A simple website was developed using **Flask** to present the Tableau dashboard in a structured format. The website provides an overview of the project along with the embedded interactive dashboard that allows users to explore electric vehicle data easily.

The website helps users analyze EV brands, efficiency, speed, and price comparisons through an interactive interface.

Website Link: *(Paste your website link here)*

GitHub Repository: *(Paste your GitHub repo link here)*

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8. Technologies Used

• **Tableau** – Data visualization and dashboard creation

- **Python** – Backend development
- **Flask** – Web framework used to host and display the dashboard
- **HTML & CSS** – Frontend structure and styling of the website
- **GitHub** – Version control and project repository management

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9. Conclusion

This project demonstrates how **data visualization techniques can be used to analyze electric vehicle datasets and identify important patterns related to battery capacity, efficiency, speed, and price**.

By integrating **Tableau dashboards with a Flask-based web application**, the project presents EV data in an interactive and easy-to-understand format. The analysis provides insights into the performance and availability of different electric vehicle brands, helping users better understand the evolving electric vehicle ecosystem.